Number theory (Stallings, 6th ed., Chapter 8)

- Prime numbers

- important for the construction of finite fields.
- " " " " many cryptographic algorithms
  e.g RSA algorithm, AES
- What are their properties?
- How can we find one when we need it?

Note: We only consider non negative integers.

- "Fundamental Theorem of Arithmetic"

Any integer $a \geq 1$ can be factored uniquely as $a = p_1^{a_1} p_2^{a_2} \cdots p_t^{a_t}$ and each exponent is a positive integer

e.g $12 = 2^2 \times 3^1$

We can express this slightly differently:

Let P be the set of all prime numbers

Then $a \geq 1$ can be written uniquely as $a = \prod_{p \in P} p^{a_p}$ where $a > 0$

where $a_p \geq 0$

$\Rightarrow$ product of all primes in the universe (but a lot of them will have 0 exponents).

Let $c = a \cdot b = \left(\prod_{p \in P} p^{a_p} \right) \left(\prod_{p \in P} p^{b_p} \right)$

Prime Numbers

$P > 1$ is a prime iff its only divisors are 1 & P.

" fundamental theorem of arithmetic"

any integer $a \geq 1$ can be factored in a unique way
a?

$a = P_1^{a_1} P_2^{a_2} \cdots P_k^{a_k}$ where $P_1 < P_2 < \cdots < P_k$

a_i is pos int
?

E.g $12 = 4 \times 3 = 2^2 \times 3^1$
$3600 = 2^4 \times 3^2 \times 5^2$

can express it slightly differently

Let P be the set of all prime ?

then any $a > 1$ can be written as $a = \prod P^{a_P}$

Usefullness:

Let $c = a \cdot b = \left(\prod_{p \in P} p^{a_p}\right) \cdot \left(\prod_{p \in P} p^{b_p}\right)$

but we know that $c = \prod_{p \in P} p^{c_p}$

$\therefore c_p = a_p + b_p$ for all $p \in P$

* more useful than that?
$\Rightarrow$ if $a | b$ then $a_p \leq b_p$ for all p
b divides evenly in a
(no remainder)

if $g = \gcd(a, b)$, then $g_p = \min(a_p, b_p)$ for all p.

eg $300 = 2^2 \times 3^1 \times 5^2$

$18 = 2^1 \times 3^2$ $\qquad \gcd(300, 18) = 2^1 \times 3^1 \times 5^0$
$= 6$

its faster than Euclid but requires that we know the factorization for a & b.

$\Downarrow$

but we know that $c = \prod_{p \in P} p^{c_p}$

$\therefore$ $c_p = a_p + b_p$ for all $p \in P$

- if $a \mid b$ then $a_p \leq b_p$ for all p.
- if $d = \gcd(a, b)$ then $d_p = \min(a_p, b_p)$ for all p

ex : $300 = 2^2 \times 3 \times 5^2$

$18 = 2^1 \times 3^2$

$\gcd(300, 18) = 2^1 \times 3^1 \times 5^0 = 6$

$p \nmid a$

- Fermat's (little) Theorem (and proof)

If p is prime and a is a positive integer not divisible by p, then $\boxed{a^{p-1} \bmod p \equiv 1 \bmod p}$

Proof: Consider the set $S = \{1, 2, \ldots, p-1\}$ multiply each element by a mod p to get

$X = \{a \bmod p, 2a \bmod p, \ldots, (p-1)a \bmod p\}$ we have 2 sets, S & X

Take every element in set X mult by a reduce mod p.

None of the elements of X is zero because p does not divide a. Furthermore, no two elements of X are equal.

if p doesn't div a, it cannot divide any of those elements together.

Uniqueness (all elems in X are unique)

Assume 2 elems are the same in X mod p

(If $ja \equiv ka \bmod p$, then $j \equiv k \bmod p$ since a is relatively prime to p and can be divided out.

Contradiction But j and k are both positive integers $< p$ and so cannot be congruent mod p)

now we know S & X are just permutations of each other, what can we do with that?

* Every element of X is less than p.

$\therefore$ X is the set of integers $\{1, 2, \ldots, p-1\}$ in some order. (i.e. X is a permutation of S)

Take the product of the elements in S & X and reduce mod p. These products must be equal.

those are the same elements but just in a different order

$[1 \times 2 \times \cdots \times p-1] \bmod p = [a \times 2a \times \cdots (p-1)a] \bmod p$

$(p-1)! \bmod p = a^{p-1} (p-1)! \bmod p$

We can divide out $(p-1)!$ because it is relatively prime to p.

Can write in more a general way

$\therefore$ $\boxed{a^{p-1} = 1 \bmod p}$

Fermat's little theorem: for any integer a & p prime:

$$a^p \equiv a \pmod p$$, a doesn't have to be relatively prime to p in this way

- Euler totient function, Euler's Theorem ← generalization of Fermat's theorem

$\phi(n) \triangleq$ (equal by definition)

($\triangleq$) the number of positive integers less than n and relatively prime to n

(# of integers in the range between $1 \le k \le n$, s.t $\gcd(n,k)=1$. note: $\phi(1)$ is therefore equal to 1.)

e.g $\phi(14) = |\{1, 3, 5, 9, 11, 13\}| = 6$

Note: $\phi(1)$ has no meaning but is defined to have value 1.

For prime p: $\phi(p) = p - 1$

Let n be the product of 2 primes p & q with $p \neq q$. The set of positive integers less than n is $\{1, 2, 3, \dots, pq-1\}$.

The integers in this set that are not relatively-prime to n are: $\{p, 2p, 3p, \dots, (q-1)p\}$ ($n = p \cdot q$)

and: $\{q, 2q, 3q, \dots, (p-1)q\}$

$$\therefore \phi(n) = (pq-1) - [(q-1)+(p-1)]$$
$$= pq - 1 - q - p + 2$$
$$= pq - p - q + 1$$
$$= (p-1)(q-1) = \phi(p)\,\phi(q)$$

When n is prime ⟹ Euler's theorem is just Fermat's little thm.

Euler's theorem: For every a & n that are relatively prime, $\gcd(a,n)=1$

$a^{\phi(n)} \equiv 1 \bmod n$

$\Rightarrow a^{\phi(n)+1} \equiv a \bmod (n)$

Proof is identical to Proof for Fermat's Theorem, generalizes (working for any n), n can be composite.

- Finding prime numbers

- essential for many cryptographic algorithms.
- no simple & efficient method known for doing this

$\Rightarrow$ Selecting a large number, and testing to see if it is prime.

Miller-Rabin algorithm

Note: For any positive odd integer n, $\boxed{n \geq 3}$, $n > 2$

$n-1 = 2^k q$ with $k > 0$ and q odd

$n-1$ is even

- Two properties of primes

Property 1 - If p is prime and a is a positive int. $< p \Rightarrow$ then $a^2 \bmod p = 1$ iif $a = 1$ or $a = p-1$ → a is either 1 or -1.

$(-1)^2 \bmod p = 1$

Property 2 - Let p be a prime > 2 [$\therefore p-1 = 2^k q$]

Let a be any integer in range $1 < a < p-1$.

From Fermat's theorem, $a^{p-1} \bmod p = 1$

$\therefore a^{2^k q} \bmod p = 1$

Consider the following sequence: $a^q \bmod p$, $a^{2q} \bmod p$, $a^{4q} \bmod p$, ..., $a^{2^{k-1}q} \bmod p$, $a^{2^k q} \bmod p$

The last number has value 1. Furthermore, each number in the seq is the square of the previous number. $\therefore$ Either the first number in the sequence is 1

What does that mean?

and so all numbers are 1 or some number is not 1 but its square is 1 i.e. that number must be $p-1$

Cons — Miller-Rabin (Algo or test) works as follows:

Given an odd integer n (>2) (some value) ($\therefore n-1=2^k q$)

If n is prime then either $a^q \bmod n = 1$ or some value in ($a^q \bmod n$, $a^{2q} \bmod n \dots, a^{2^{k-1}q} \bmod n$) equals $n-1$ (satisfies)

Note: The inverse is not necessarily true.

Test (n):

1. Find k & q with $k>0$ and q odd such that $n-1=2^k q$
2. Select random integer a, $1<a<(n-1)$.
3. If $a^q \bmod n = 1$: return "inconclusive"
4. for $j=0$ to $k-1$ do
5. if $a^{2^j q} \bmod n = n-1$ return "inconclusive"
6. return ("composite") — it discovers that it is composite

prime $\rightarrow$ $a^q \bmod n = 1$ or $a^{2^k q} \bmod n = n-1$

$\longleftarrow$ we are using this direction for testing.

Prob $< \frac{1}{4}$ (inconclusive for a composite n)

$\therefore$ try t different times (ie different orders of a).

How often do we need to run the test on a number n?

It can be shown that for an odd int. n that is not prime and a randomly chosen a with $1 < a < (n-1)$, the probability that the test returns inconclusive $< \frac{1}{4}$

$\Rightarrow$ For t different values of a, the probability that the test will return "inconclusive" for all of them $< \left(\frac{1}{4}\right)^t$

e.g for $t = 10$, prob. is less than 1 in a million.

$\therefore$ Run the Miller-Rabin test t times with diff a for sufficiently large t. If it always returns inconclusive, then n is probably prime.

If it returns composite once $\Rightarrow$ STOP TESTING

"probabilistic primality testing algorithm"

$\Rightarrow$ can never guarantee a number is prime.

Deterministic primality testing algorithms?

E.g divide by small primes (up to $n/2$) (If none of them divides evenly into n $\Rightarrow$ n is prime)

E.g AKS (Agrawal, Kayal, Saxena) : 2002

- polynomial time algorithm for proving that a positive integer is prime

Complexity: Order $(\log^6 n)$ whereas complexity of M.R is $O(\log^3 n)$

How long until the test finds a (probable) prime?

- "Prime number theorem"
- Primes near n are spaced on average 1 every $\boxed{\ln n}$ integers

$\therefore$ need to test $\ln n$ integers on average in order to find a prime (actually: $\frac{1}{2}\ln n$ as we can get rid of even numbers).

e.g prime near 2^{200}: test $\frac{1}{2}\ln(2^{200}) \approx 69$ integers.

For each of these 69 integers, we are using the Miller-Rabin test.

How many primes are there?

- If the primes are spaced $\ln n$ apart (on average), then for a 200-bit number (i.e. around 2^{200}). They would be spaced about every 140 integers.

But how many 200-bit odd integers are there?

$\Rightarrow 2^{199}$. If about 1 out of 140 $(\sim \frac{1}{2^7})$ is a prime, then there are roughly 2^{192} 200-bit primes

$\Rightarrow$ If you pick a prime at random, an attacker is not likely to guess which one you've picked.